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Key Techniques and Challenges in NeRF-Based Dynamic 3D Reconstruction

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Abstract:

This paper explores the key techniques and challenges in dynamic scene reconstruction with neural radiance fields (NeRF). As an emerging computer vision method, the NeRF has wide application potential, especially in excelling at 3D reconstruction. We first introduce the basic principles and working mechanisms of NeRFs, followed by an in-depth discussion of the technical challenges faced by 3D reconstruction in dynamic scenes, including problems in perspective and illumination changes of moving objects, recognition and modeling of dynamic objects, real-time requirements, data acquisition and calibration, motion estimation, and evaluation mechanisms. We also summarize current state-of-the-art approaches to address these challenges as well as future research trends. The goal is to provide researchers with an in-depth understanding of the application of NeRFs in dynamic scene reconstruction, as well as insights into the key issues faced and future directions.

Keywords: neural radiance fields; 3D vision; dynamic scene reconstruction

1 Introduction

Dynamic 3D reconstruction is an important research topic in the field of computer vision, and its applications cover a wide range of fields, such as virtual reality, medical imaging, and industrial automation^[1-4]. Dynamic scenes typically involve moving objects or environments, and 3D reconstruction algorithms for such scenes primarily concentrate on reconstructing non-rigid objects. This often includes addressing general deformations, joint motions, and the capture and reconstruction of human movements. The challenges of dynamic scene 3D reconstruction can be subdivided into related subproblems including motion estimation, feature extraction and matching, data alignment and fusion, motion removal, and segmentation. These subproblems are intricately interconnected. In recent years, as static scene 3D reconstruction algorithms have matured, research on algorithms for reconstructing dynamic scenes has emerged as a prominent and challenging area of focus. Dynamic 3D reconstruction techniques based on the neural radiance field (NeRF) have attracted extensive attention^[5-9]. As an

emerging computer vision method, the powerful potential of NeRFs is fully demonstrated in 3D scene reconstruction^[10–14]. The purpose of this paper is to deeply explore the key techniques and challenges in 3D reconstruction based on NeRFs.

First, we introduce the fundamentals and working mechanisms of NeRFs to provide researchers with a foundational understanding. NeRFs draw on the ideas of deep learning and neural networks and apply them to 3D reconstruction tasks, bringing new possibilities to dynamic 3D reconstruction by learning the ability to recover 3D information from multi-view images^[15–22].

This is followed by an in-depth discussion of the key technical challenges in performing 3D reconstruction in dynamic environments. These challenges include, but are not limited to, viewpoint and illumination variations of moving objects, object identification and modeling, real-time requirements, data acquisition and calibration challenges, the complexity of motion estimation, and effective evaluation mechanisms for reconstruction results. These issues are the core challenges in the dynamic 3D reconstruction and require in-depth research and innovative solutions.

Finally, we summarize the current state-of-the-art approaches and trends to address these challenges, and look ahead to future research directions. By exploring these key techniques and challenges, we are expected to promote further development in dynamic 3D reconstruction, and provide support and insights for the realization of more accurate, efficient and widely used 3D reconstruction techniques.

2 Scene Reconstruction with NeRFs

2.1 NeRFs

In 3D reconstruction, NeRFs, as an implicit representation technique, can depict 3D models through implicit functions learned by neural networks. This method has valuable applications in areas like image generation, viewpoint generation, and re-illumination. This section begins by summarizing and introducing the methods that utilize neural networks as implicit representations for scene geometry, before presenting the concept of NeRFs. A prevalent technique for employing neural networks to implicitly convey 3D geometry is the occupancy network^[23–24]. This approach employs a neural network to predict the binary occupancy of each point in space, essentially training a binary classification network for 3D space, as illustrated in Fig. 1. The key advantage of this method lies in its use of continuous functions to articulate 3D space. In comparison to prior approaches such as voxels and meshes, it excels in describing complex geometric shapes without necessitating additional spatial storage.

Apart from directly classifying space into two categories based on model existence, there exists another implicit representation method that portrays the 3D model through the regression of a signed distance function (SDF)^[25–26]. This approach allows for the continuous representation of 3D models, enabling the modeling of even those with intricate topologies.

Building upon the SDF method, researchers have enhanced and applied it to represent models with intricate details. One notable example is the Pixel Aligned Implicit Function (PIFu) method^[26], which learns the details of a 3D model by projecting spatial points onto a pixel-aligned feature space, enabling high-resolution reconstruction, e.g., of a dressed human model. However, these methods often rely on known 3D shapes as supervisory information, which is challenging to obtain in many applications. Consequently, subsequent research has aimed to relax this condition by directly utilizing images as supervision. Some studies introduced differentiable drawing techniques, incorporating drawing steps into neural networks to train the network based on errors in image rendering. NIEMEYER et al.^[27] employed

a placeholder network as the representation structure for 3D model geometry, determining ray-model surface intersection points using numerical methods. Each intersection point served as input for the neural network to predict the corresponding color value. SITZMANN et al.^[28] predicted color and feature vectors for each 3D spatial coordinate, proposing a differentiable drawing function composed of recurrent neural networks to locate object surface. However, these methods often struggled with complex shapes, limited to handling simple structures with low geometric complexity, yielding overly smooth drawing results. Against this backdrop, MILDENHALL et al.^[29] introduced NeRF, a novel representation method that uses only input images as supervisory information. NeRF can accurately fit implicit functions for high-resolution geometric shapes, achieving photo-realistic viewpoint synthesis results for complex scenes. The overall process of this algorithm is depicted in Fig. 1. NeRF employs a multi-layer perceptron to express a 5D vector function, describing both geometric and color information of a 3D model.

NeRF relies solely on input images as supervisory information. This innovative approach excels at fitting precise implicit functions in high-resolution geometric shapes, consequently attaining photo-realistic viewpoint synthesis results for complex scenes. NeRF represents the 3D scene as a differentiable and continuous radiation field F_θ :

$$F_\theta(\mathbf{x}, \mathbf{d}) = [\sigma, \mathbf{c}] \quad (1),$$

where $\mathbf{x} = (x, y, z)$ denotes the coordinates of a point in 3D space; $\mathbf{d} = (d_x, d_y, d_z)$ denotes the normalized viewing direction; θ is the set of variables that parameterize the model. e.g., an multilayer perceptron (MLP); σ denotes the density estimate at the point $\mathbf{x}$, which is the probability that the ray terminates at the point. Assuming that the position of the current camera center is $\mathbf{o} \in \mathbb{R}^3$ and connecting any pixel on the image with the center, we can get the view direction $\mathbf{d} \in \mathbb{R}^3$. We parameterize a ray extending from the camera center $\mathbf{o}$, with the view direction $\mathbf{d}$ as follows:

$$\mathbf{l}(t) = \mathbf{o} + t\mathbf{d}, t \in (-\infty, +\infty) \quad (2).$$

According to the formula of the volume rendering, the color value of the pixel can be expressed as

$$\mathbf{C} = \int_{t_n}^{t_f} T(t) \sigma(\mathbf{l}(t)) \mathbf{c}(\mathbf{l}(t), \mathbf{d}) dt \quad (3),$$

where

$$T(t) = \exp\left(-\int_{t_n}^t \sigma(\mathbf{l}(s)) ds\right) \quad (4).$$

The transmittance, as defined in Eq. (4), quantifies the probability that a ray, traveling between points t_n and t , is absorbed, scattered, or reflected by objects encountered along its path.

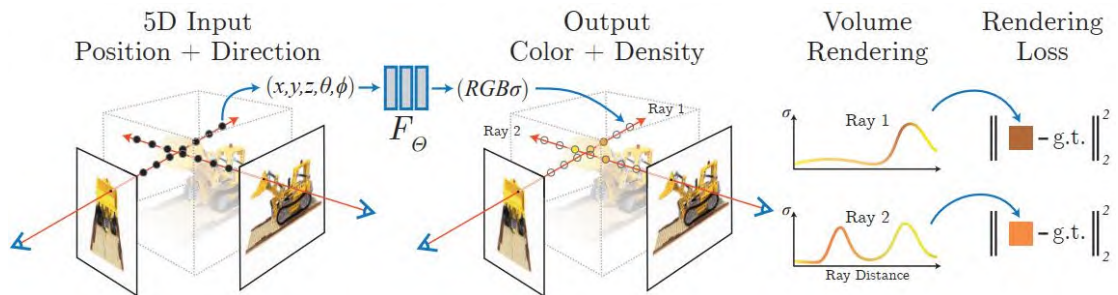


Figure 1. An overview of NeRF scene representation and differentiable rendering procedure

Thanks to the differentiable volume rendering process, it can be seamlessly integrated into the training of the aforementioned neural network. This enables a training process that relies exclusively on image color values as supervision. Furthermore, to prevent the loss of high-frequency information in the synthesized image, NeRF employs positional encoding on the input variables^[30]. This process involves mapping the variables to their Fourier features. Inspired by positional encoding techniques employed in natural language processing, such as those used in Transformers, NeRF utilizes a similar approach to encode input coordinates. This involves using a set of basis functions, which can either be fixed or learned^[31]. The spatial embeddings generated by these basis functions simplify the task of the MLP in learning the mapping from a location to a specific value, as the input space is effectively partitioned. The positional encoding method used in NeRF is defined as follows:

$$\mathbf{x} \mapsto [\cos(\mathbf{M}\mathbf{x}), \sin(\mathbf{M}\mathbf{x})] \quad (5),$$

where

$$\mathbf{M} = [\mathbf{I} \quad 2\mathbf{I} \quad 2^2\mathbf{I} \quad \dots \quad 2^{p-1}\mathbf{I}]^\top \quad (6).$$

Here, $\mathbf{x}$ represents the input coordinate, and p stands for a hyperparameter that governs the frequencies utilized, with its value dependent on the target signal resolution. The “soft” binary encoding of the input coordinates is employed, facilitating the network’s access to higher frequencies within the input.

2.2 Dynamic Scene Radiance Field Reconstruction

The initial work on NeRF was only for static scenes; thus one of the most important branches of the improved NeRF is the construction of radiation fields on dynamic scenes, which are closely related to realistic scenes^[29, 32–33]. Dynamic scene NeRFs are 3D scene representations learned from a set of posed images. They are formulated to address the challenge of rendering photo-realistic images from unseen viewpoints, and adopt implicit representation based on coordinates, which then maps spatial points to density and color^[34–35]. Recent research in this field is extensive. Based on the different reconstruction objects, we elaborate on them from human-based and scene-based reconstruction perspectives, as shown in Table 1.

1) Human-based Reconstruction

The dynamic 3D reconstruction of the human body is, to a certain extent, associated with the application requirements of remote presentation, virtual reality, augmented reality, virtual fitting, and similar domains. PENG et al.^[36] presented a novel method, named Neural Body, for dynamic human 3D reconstruction. This approach introduces a neural mixed weight domain to generate a deformation field, combining the mixed weight field with 3D human bones to achieve commendable results in dynamic 3D reconstruction of the human body. However, it should be noted that this method relies on bone-driven deformation templates, which exhibit limited universality and necessitate considerable time for reconstruction. Particularly, in cases involving non-rigid deformations with intricate clothing, the reconstruction effectiveness tends to be suboptimal.

Similar to the Neural Body method, Neural Actors (NA)^[37] and hybrid volumetric-textural rendering (HVTR)^[38] use the skinned multi-person linear model (SMPL) to represent deformation states. They utilize proxies to explicitly carve out the surrounding 3D space into the canonical pose embedded in NeRF. To facilitate the recovery of high-fidelity details in both geometry and appearance, they employ additional 2D texture maps defined on the SMPL surface as additional conditioning to the NeRF MLP. CAI et al.^[39] introduce a template-free method termed neural dynamic reconstruction (NDR), which proficiently reconstructs dynamic scenes from monocular videos. This method leverages color and depth information to optimize surface deformation and employs a neural invertible network to ensure cyclic

consistency between any two frames. Additionally, topology-aware networks are employed to model topology variables, effectively addressing challenges related to topology changes. Nonetheless, it's worth noting that the NDR method exhibits subpar reconstruction performance for dynamic scenes with rapid motion and demands a substantial number of computing resources. Another method, named HumanNeRF^[40], demonstrates how to train a NeRF for a specific participant based on monocular input data, utilizing a skeleton-driven motion field refined by a general non-rigid motion field. CHEN et al.^[41] propose an effective general framework called the generic model-based neural radiance field (GM-NeRF) for synthesizing free-viewpoint images. Specifically, they first register the appearance codes of multi-view 2D images onto a geometric proxy through a geometry-guided attention mechanism. This helps mitigate the misalignment between inaccurate geometric priors and the pixel space. Building upon this, they further perform neural rendering and partial gradient backpropagation to achieve efficient perceptual supervision and enhance the perceptual quality of the synthesis.

While the aforementioned approaches demonstrate promising results in portrait scenarios, their applicability diminishes when dealing with highly non-rigid deformations, particularly in the context of articulated motion of a human body captured from a single view. Therefore, methods leverage the human skeleton embedding explicitly. The Neural Articulated Radiance Field (NARF)^[42] is trained using pose-annotated images. Joint objects are decomposed into multiple rigid object parts, with their local coordinate systems and global shape variations located at the top. The converged NARF can be used to render novel views by manipulating poses, estimate depth maps, and perform body part segmentation. In contrast, A-NeRF^[43] learns actor-specific volumetric neural body models in a self-supervised manner from a single monocular camera. This method combines dynamic NeRF volumes with the explicit controllability of articulated human skeletons and reconstructs poses and radiance fields through a comprehensive analysis approach. Once trained, the radiance field can be used for novel viewpoints synthesis and motion retargeting. They demonstrate the benefits of using the learned non-surface model, which enhances the accuracy of human pose estimation in monocular videos through photometric reconstruction loss. A-NeRF is trained on monocular videos, while Animatable NeRFs (ANRF)^[44] is a skeleton-driven approach used for reconstructing human body models from multi-view videos. Its core component is a novel motion representation called the neural blend weight field, which is combined with the 3D human skeleton to generate a deformation field. Similar to several general non-rigid NeRF approaches, ANRF maintains a canonical space and estimates bidirectional correspondences between multi-view inputs and canonical frames. The reconstructed animatable human body model can be used for free-viewpoint rendering and re-rendering under new poses. Additionally, by running a marching cube on the volume density of discretized canonical space points, it is possible to extract a human body mesh from ANRF. The method achieves high visual accuracy for the learned human body model, and the authors suggest addressing complex non-rigid deformations on observed surfaces, such as those caused by looseness. The authors recommend future work to improve the handling of complex non-rigid deformations on observed surfaces, such as those caused by loose clothing.

2) Scene-based reconstruction.

DU et al.^[45] proposed NeRFlow to learn dynamic 4D spatiotemporal scenes. NeRFlow consists of two separate modules: a radiation field (top) trained by neural rendering, and a flow field (bottom) trained by 3D keypoint correspondence. The two fields are then kept consistent, which enables the radiation field to acquire prior information from earlier states. SONG et al.^[46] used a feature flow approach to model dynamic radiation scenes. The author mainly used time-dependent sliding windows for points in 4D space to generate flow features, and then decomposed the dynamic scene into predicted static,

deformation and new scene decomposition fields via a point-by-point probabilistic method. Finally, the expectation of the decomposition fields was fed into NeRF for modeling. However, since local feature channels were used to model each frame in the scene, which enables streaming but limits the representation of remote repetitive activities, it might be used multiple times to reconstruct the same action, resulting in a waste of time. GAO et al.^[47] proposed DynamicNeRF, an algorithm for generating novel views from any viewpoint of a monocular dynamic scene video and any input time step. The algorithm takes a monocular video with N frames and a binary mask of the foreground object for each frame as input, and models the time-varying structure and the appearance of the scene using continuous and differentiable functions.

However, some authors believe that the key problem in solving dynamic scene rendering lies in the encoding of temporal information. FANG et al.^[48] proposed TiNeuVox, which used a combination of optimizable explicit voxel features and temporal information encoding to quickly generate dynamic scenes. They first input the point coordinates and temporal coding into a deformation network to obtain the offset coordinates, then interpolated the voxels according to the offset coordinates to obtain the voxel features, and finally connected the original coordinates, temporal coding, and voxel features and fed them into a NeRF network to obtain the colors and densities. However, they did not consider the relationship between neighboring frames, so there is a slight problem with the coherence of the video. In addition, ABOU-CHAKRA et al.^[49] introduced an online method for generating dynamic scenes. Inspired by particle dynamics, they proposed a new particle coding that enables the intermediate features of NeRF to move in conjunction with the geometry they represent. As a result, the authors have achieved automated generation of dynamic scenes by back-projecting rendering losses to particle positions and encoding particle parameters.

Another class of methods introduces additional deformation fields to predict the motion of points by mapping their coordinates to a normative space where large motion or geometric changes can be captured and learned. PUMAROLA et al.^[50] proposed a method to extend NeRFs to the dynamic domain, D-NeRF, which allows a single camera to reconstruct and draw a new image as it moves under both rigid and nonrigid motion images of the scene. Therefore, it is necessary to include time as an additional input to the system and to divide the learning process into two main phases: one phase encodes the scene into a canonical space and the other maps this canonical representation to a deformed scene at a specific time.

Other methods improve dynamic neural rendering in various ways, e.g., distinguishing between foreground and background. TRETSCCHK et al.^[51] proposed non-rigid NeRF (NRNERF), a reconstruction and new view synthesis method for general non-rigid dynamic scenes. The method takes an RGB image of a dynamic scene (e.g., from monocular video recordings) as input and creates high-quality representations of spatio-temporal geometry and appearance. Meanwhile, quality enhancement using depth information can improve dynamic neural rendering. ATTAL et al.^[52] noted that neural networks can represent and accurately reconstruct the radiance field of a static 3D scene (e.g., NeRF). However, dynamic scene approaches for monocular video capture rely on data-driven priors to reconstruct dynamic content. To address this, the authors replaced this a priori information with time-of-flight (TOF) camera measurements and introduced a neural representation based on a continuous-wave TOF camera image formation model. Instead of using processed depth maps, the method models the raw TOF sensor measurements to improve the reconstruction quality and to avoid the problems of low reflectivity regions, multipath interference, and limited explicit depth range of the sensor. Additionally, setting keyframes to produce sharper results is another effective approach. LI et al.^[53] proposed a new

3D video synthesis method that compactly and expressively represents multi-viewpoint video recordings of dynamic real scenes, allowing for high-quality viewpoint synthesis and motion interpolation.

The state-of-the-art method based on temporally varying NeRFs, also known as Dynamic NeRFs, has demonstrated impressive results in this task. However, for long videos with complex object motions and uncontrolled camera trajectories, the method may result in blurry or inaccurate renderings. To address this issue, LI et al.^[54] proposed a novel approach. Instead of encoding the entire dynamic scene within the weights of an MLP, this method employs a volume-image-based rendering framework. This framework synthesizes new viewpoints by aggregating features from nearby views in a scene-motion-aware manner, overcoming these limitations. The system retains the capability of previous methods to model complex scenes and view-dependent effects. Still, it can also synthesize realistic new views for long videos with complex dynamic scenes and unconstrained camera trajectories.

Table 1. An overview of the discussed methods in Section 2.2

Object	Method	Data Attribute	Required Data	3D Representation	Year
Human-based	Neural body ^[36]	Multi-view	I+P+S	V	2021
	Neural actors ^[37]	Multi-view	I+P+S	P+VD	2021
	HVTR ^[38]	Multi-view	P+S	V	2022
	NDR ^[39]	Monocular	I+P	P+VD	2022
	HumanNeRF ^[40]	Multi-view	I+P	P+VD	2022
	GM-NeRF ^[41]	Multi-view	I+P+S	P+VD	2023
Scene-based	NeRFlow ^[45]	Multi-view	I+P	P+VD	2021
	NeRFPlayer ^[46]	Multi-view	I+P	V	2023
	DynamicNeRF ^[47]	Monocular	I+P+M	P+VD	2021
	TiNeuVox ^[48]	Multi-view	I+P	V	2022
	NRNeRF ^[49]	Monocular	I+P	V	2022
	D-NeRF ^[50]	Multi-view	I+P	P+VD	2021
	NRNeRF ^[51]	Monocular	I+P	V	2022
	Torf ^[52]	Multi-view	I+P	P+VD	2022
	Neural 3d ^[53]	Multi-view	I+P	P+VD	2022
	DynIBaR ^[54]	Monocular	I+P	P+VD	2023

GM-NeRF: generic model-based neural radiance field

HVTR: hybrid volumetric-textural rendering

NDR: neural dynamic reconstruction

NRNeRF: non-rigid neural radiance field

Notes: I: Images, P: camera poses (exact or approximate); M: object masks; P: 3D position; S: SMPL priori model; V: neural volumetric; VD: 2D viewing direction

3 Database and Evaluation

3.1 Common Database

We present the common database in this section in Table 2.

1) DNA-Rendering^[55], a large-scale, high-fidelity repository of neural actor rendering represented

by neural implicit fields of human actors. This dataset has collected data from 500 individuals, with 527 distinct sets of clothing, 269 types of daily actions, and 153 types of special performances, including relevant interactive objects for some actions. Additionally, a professional multi-view system is constructed to capture data, which contains 60 synchronous cameras with a max resolution of 4096×3000 and a frame rate of 15 frames per second.

2) ZJU_MoCap^[36]. This dataset captures nine dynamic human videos using a multi-camera system that has 21 synchronized cameras. All sequences have a length between 60 to 300 frames. In these videos, humans perform complex motions, including twirling, Taichi, arm swinging, warmup, punching, and kicking.

3) ENeRF-Outdoor^[56], a dynamic dataset of multi-purpose outdoor scenes, collected by 18 synchronized cameras. Each sequence generally has about 1 000 frames and complex motions.

4) NVIDIA^[57]. This dataset collects dynamic scenes using two methods. a) Moving monocular camera: Short-term dynamic events (about 5 s) are captured by a hand-held monocular moving camera (Samsung Galaxy Note 10) with a framerate of 60 Hz and a resolution of 1920×1080 . Sequences are subsampled if the object motion is not salient, and therefore, the degree of the scene motion is significantly larger than that of the camera's ego motion, making quasi-static dynamic reconstruction inapplicable. Four dynamic scenes are captured, which includes human activity, human-object interactions, and animal movements; b) Stationary multi-view cameras: Eight scenes are captured by a static camera rig with 12 cameras (GoPro Black Edition).

Table 2. Information on commonly used datasets for dynamic 3D reconstruction

Name	Object	Cases	Cameras	Resolution	Year
DNA-Rendering	Human-based	439	60	4K	2023
ZJU_MoCap	Human-based	9	23	1K	2021
ENeRF-Outdoor	Scene-based	8	18	4K	2022
NVIDIA	Scene-based	12	4 scenes by monocular; 8 scenes with 12 cameras	960×540	2020

3.2 Evaluation Metrics

The synthesis of novel views through NeRF employs visual quality assessment metrics as benchmarks. These metrics aim to evaluate the quality of individual images with (full-reference) or without (no-reference) ground truth images. To date, peak signal-to-noise ratio (PSNR), structural similarity index measure (SSIM)^[58], and learned perceptual image patch similarity (LPIPS)^[59] are the most commonly used metrics in NeRF related literature.

1) **PSNR** is one of the important metrics for measuring image quality. The formula for calculating PSNR is as follows:

$$\text{PSNR} = 10 \cdot \log_{10} \left(\frac{\text{MAX}^2}{\text{MSE}} \right) \quad (7),$$

where MAX is the maximum possible range of pixel values in the image (usually 255 for 8-bit images), and MSE is the average of squared differences between corresponding pixels. A higher PSNR value indicates better image quality, making it a widely used standard for evaluating image reconstruction quality in image processing and compression. It is important to note that PSNR may not fully align with

human perception of image quality. Therefore, in certain applications, other metrics such as SSIM or LPIPS are employed to more comprehensively assess image quality.

2) **SSIM** consists of three contrast similarity modules, namely: luminance, contrast and structure.

Luminance modules can be written as:

$$l(x, y) = \frac{2\mu_x\mu_y + C_1}{\mu_x^2 + \mu_y^2 + C_1} \quad (8),$$

where μ_x and μ_y are the average gray values of images I_x and I_y , respectively; C_1 is a constant.

Contrast modules can be written as:

$$c(x, y) = \frac{2\sigma_x\sigma_y + C_2}{\sigma_x^2 + \sigma_y^2 + C_2} \quad (9),$$

where σ_x and σ_y are the standard deviation of images I_x and I_y , respectively; C_2 is a constant.

Structure modules can be written as:

$$s(x, y) = \frac{\sigma_{xy} + C_3}{\sigma_x\sigma_y + C_3} \quad (10).$$

Finally, SSIM can be formulated as

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \quad (11).$$

3) **LPIPS**. LPIPS distance is used to measure the average feature distances between two images, which is calculated from the weighted pixel-level MSE of the multilayer feature maps.

$$\text{LPIPS}(x, y) = \sum_l \frac{1}{H_l W_l} \sum_{h,w}^{H_l W_l} \|w_l \square (x'_{hw} - y'_{hw})\|_2^2 \quad (12),$$

where x'_{hw} and y'_{hw} are the features of the reference and evaluation images at pixel width w , pixel height h , and layer l . H_l and W_l are the height and width of the feature maps of the corresponding layers.

4 Challenges

3D reconstruction in dynamic environments involves a series of complex and important technical challenges. One is perspective and illumination changes of moving objects. In an ever-changing dynamic scene, the positions and orientations of objects are constantly changing over time, and thus their appearance and perspective change significantly at different moments. In addition, lighting conditions may vary constantly across time and space, further complicating the accurate capture and modeling of moving objects. Recognition and modeling is another challenging area. The need to reliably identify and track multiple moving objects in dynamic environments requires highly accurate object recognition and modeling techniques to efficiently handle complex scenes and ensure the accuracy and consistency of 3D models.

Data acquisition can be challenging in dynamic environments where the position and orientation of moving objects may change over time. Therefore, it is important to address the complexity of data acquisition and ensure the accuracy and consistency of the sensors. Another major problem faced in dynamic 3D reconstruction compared with static 3D reconstruction is the need to accurately estimate camera and object motion. Finally, in order to validate and evaluate the quality of 3D reconstruction results, it is necessary to develop evaluation methods and metrics applicable to dynamic environments, especially for monocular dynamic scene reconstruction. How to evaluate the quality of image reconstruction from different viewpoints at the same moment is a very urgent problem in engineering practice. Comprehensive consideration of these technical challenges and continuous improvement of algorithms and methods are the key to realize high-quality 3D reconstruction in dynamic environments.

Fig. 2 shows a 3D reconstruction process for dynamic and static distributed scenes, which is also the basic process used in practice. Taking this process as an example, we will specifically introduce the key issues and challenges in dynamic 3D reconstruction. When the input is a video stream from a common capture device, extensive preprocessing is required. The video stream is first decomposed into video frames, and then the existing or improved instance segmentation algorithms are applied to RGB image of these frames to generate static and dynamic masks. Due to the input and limitations of the existing terminal memory and graphics, we need to reasonably select key frames from the full set of video frames, including the key information of the dynamic scene, continuous changes in motion, significant changes in illumination, accurate camera viewpoints, frames with overlapping regions, the depth and optical flow of the correctly calculated. These effective and highly accurate information can maximize the quality of 3D reconstruction of the dynamic scene. The effective high-precision frame information can maximize the quality and accuracy of the 3D reconstruction of the dynamic scene and help produce better results.

The obtained information is then fed into the neural network for processing. In static NeRF, only the position information is input to derive static color features and density features. Since a dynamic scene exhibits two different attributes, static and dynamic, at the same sampling point under different viewpoints and at different times, the static output is used as part of the input to the dynamic neural network. This network is trained with the spatiotemporal information, thereby constraining the overall convergence. The team is constantly conducting experiment, and one of the key challenges lies in the sparsity of the dynamic data, which makes it difficult to converge to high precision results. Therefore, there is an urgent need to continuously seek additional constraints and methods to improve modeling precision and accuracy. These constraints can cover a number of aspects, including but not limited to depth constraints, temporal continuity, motion modeling, optical flow coherence, and multi-sensor fusion. By introducing these constraints, we hope to make a bigger breakthrough in reconstruction quality, which in turn will enable more accurate capture and reproduction of complex dynamic scenes. Finally, voxel rendering is performed on the dynamic and static data obtained from training, in order to complete the model reconstruction based on NeRF rendering and to generate new viewpoints over time.

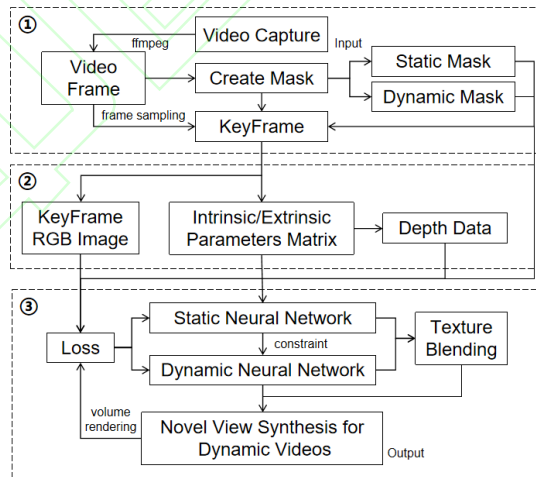


Figure 2. A novel viewpoint synthesis framework based on dynamic and static distributions

5 Development Trends

The field of 3D reconstruction with NeRF still faces a series of problems, and the following are what we consider as possible future research directions: 1) Developing robust machine learning and deep learning models with generalized modeling capabilities for dynamic scenes, including improving model robustness to handle challenges such as noise, occlusion, and incomplete data; 2) Exploring more

effective constraints and implicit modeling in which the physical and geometric properties of the scene are better captured; 3) Advancing multimodal fusion, including images, point clouds, sound, etc. This helps improve the understanding and modeling of dynamic scenes and make reconstruction results more comprehensive and accurate; 4) Promoting self-supervised learning to reduce the dependence on labeled data. Especially in the absence of large-scale labeled data, self-supervised learning methods can improve the performance of dynamic 3D reconstruction; 5) Conducting semantic modeling of dynamic scenes.

6 Conclusions

In this paper, we delve into the key techniques and challenges in the field of dynamic 3D reconstruction based on NeRFs. Dynamic 3D reconstruction is an important research direction in the field of computer vision with a wide range of applications. As an emerging computer vision method, NeRF has a strong potential in 3D scene reconstruction.

This paper first introduces the basic principles and working mechanism of NeRF, which provides readers with a foundation for understanding this novel approach. NeRF draws on deep learning and neural networks and applies them to the 3D reconstruction task to learn to recover 3D information from multi-view images. It then presents an in-depth discussion of the key technical challenges facing 3D reconstruction in dynamic environment, including viewpoint and illumination variations of moving objects, object recognition and modeling, real-time requirements, data acquisition and calibration challenges, complexity of motion estimation, and effective mechanisms for evaluating reconstruction results. These core challenges require in-depth research and innovative solutions. In addition, this paper summarizes current technology trends and approaches to address these challenges and outlines future research directions. It emphasizes the importance of directions such as robustness of machine learning and deep learning models, more efficient constraints, multimodal fusion, self-supervised learning, and semantic modeling of dynamic scenes.

Finally, this paper emphasizes that the field of dynamic 3D reconstruction will continue to thrive with the rise of the metaverse. These research directions will help to continuously improve the performance and applicability of dynamic 3D reconstruction, drive innovation and development in this field, and create more exciting applications and possibilities. We look forward to making more breakthroughs in this challenging and opportune field, and contributing more support and insight to the future of 3D reconstruction technology.

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